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Used paper recycling apparatus

Abstract

A used paper recycling apparatus includes a defibration unit that defibrates used paper to generate a defibrated product, an accumulation unit that accumulates the defibrated product to form a web, a sheet forming unit that forms a sheet by applying at least one of heating and compression to the web, a cutting unit that cuts the sheet into a predetermined length, and an optical sensor that is provided at an exit of the sheet forming unit and that measures density information of the sheet.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Patent No.	Application Date	Country	CPC
2017-008426	12/2016	JP	N/A

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-018521, filed Feb. 9, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a used paper recycling apparatus.

2. Related Art

(3) In the related art, as disclosed in JP-A-2017-8426, there is known a used paper recycling apparatus that defibrates used paper to produce new paper.

(4) However, in the used paper recycling apparatus described above, when the recycled paper is repeatedly recycled, the whiteness of the paper tends to gradually decrease, and the paper may be unsuitable for use. Therefore, it is necessary to determine whether the recycled paper is acceptable. However, there is a problem that the operation efficiency of the used paper recycling apparatus is lowered when the paper discrimination is performed by a human.

SUMMARY

(5) According to an aspect of the present disclosure, a used paper recycling apparatus includes a defibration unit that defibrates used paper to generate a defibrated product, an accumulation unit that accumulates the defibrated product to form a web, a sheet forming unit that forms a sheet by applying at least one of heating and compression to the web, a cutting unit that cuts the sheet into a predetermined length, and an optical sensor that is provided at an exit of the sheet forming unit and that measures density information of the sheet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram showing the configuration of a sheet manufacturing apparatus.
- (2) FIG. 2A is a schematic diagram showing the configuration of a detection unit.
- (3) FIG. 2B is a schematic diagram showing the configuration of a detection unit.
- (4) FIG. 3 is a flow chart showing a control method of the sheet manufacturing apparatus.
- (5) FIG. 4A is a flow chart showing a control method of the sheet manufacturing apparatus.
- (6) FIG. 4B is a flow chart showing a control method of the sheet manufacturing apparatus.
- (7) FIG. 4C is a flow chart showing a control method of the sheet manufacturing apparatus.
- (8) FIG. 4D is a diagram showing an example of the relationship between an AD value and whiteness.
- (9) FIG. 5A is a flow chart showing a control method of the sheet manufacturing apparatus.
- (10) FIG. 5B is a flow chart showing a control method of the sheet manufacturing apparatus.
- (11) FIG. 5C is a diagram showing an example of sheet discrimination.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (12) First, the configuration of a sheet manufacturing apparatus **1** as a used paper recycling apparatus will be described. The sheet manufacturing apparatus **1** dry-defibrates used paper as a raw material to fiberize the used paper, and then compresses (pressurizes), heats, and cuts the fiber to manufacture new paper (sheet S), for example. By mixing various additives into the fiberized raw material, it is possible to improve the binding strength and whiteness of the sheet S, and add functions such as color, fragrance, and flame resistance according to the application. In addition, by controlling the density, thickness, and shape of the sheet S, it is possible to manufacture the sheets S of various thicknesses and sizes, such as A4 or A3 office paper, business card paper, and the like.
- (13) As shown in FIG. 1, the sheet manufacturing apparatus **1** includes a supply unit **10**, a coarsely crushing unit **12**, a defibration unit **20**, a sorting unit **40**, a first web forming unit **45**, a rotating body **49**, and a mixing unit **50**, an accumulation unit **60**, a transport unit **79**, a sheet forming unit **80**, a cutting unit **90** and a detection unit **100**.
- (14) Further, the sheet manufacturing apparatus **1** includes a controller **300** that controls the above units and the like. The controller **300** includes a CPU, a memory, a control circuit, an I/F (interface), and the like. The CPU is an arithmetic processing unit. The memory is a storage device that secures an area for storing various programs and various parameters of the CPU or a work area, and has storage elements such as a RAM and an EEPROM. The I/F is configured to be capable of transmitting and receiving information to and from an external device such as an information processing terminal. The controller **300** causes the CPU to execute arithmetic processing according to various programs and the like, and controls respective drive units and the like through the control circuit.
- (15) The sheet manufacturing apparatus **1** also includes humidifying units **202**, **204**, **206**, **208**, **210**, and **212**, for example, for the purpose of humidifying the raw material and humidifying the space in which the raw material moves. Humidification suppresses sticking of raw materials due to static electricity. The humidifying units **202**, **204**, **206**, and **208** are, for example, evaporative or warm air evaporative humidifiers. The humidifying units **210** and **212** are, for example, ultrasonic humidifiers.
- (16) The supply unit **10** supplies raw materials to the coarsely crushing unit **12**. The raw material supplied to the coarsely crushing unit **12** may be any material containing fibers, and examples thereof include paper, pulp, pulp sheets, non-woven fabrics, cloth, and woven fabrics. A configuration in which the sheet manufacturing apparatus **1** uses used paper as a raw material is exemplified below. The supply unit **10** includes, for example, a stacker that stacks and accumulates used paper, and an automatic input device that feeds the used paper from the stacker to the coarsely

crushing unit **12**.

(17) The coarsely crushing unit **12** cuts the raw material supplied by the supply unit **10** with a coarsely crushing blade **14** into coarsely crushed pieces. The coarsely crushing blade **14** cuts the raw material in air such as the atmosphere. The coarsely crushing unit **12** includes, for example, a pair of coarsely crushing blades **14** that pinches and cuts the raw material, and a drive unit that rotates the coarsely crushing blades **14**, and can have the same configuration as a so-called shredder. The shape and size of the coarsely crushed pieces are not restricted as long as they are suitable for the defibration process in the defibration unit **20**. The coarsely crushing unit **12** cuts the raw material into pieces of paper having a size of, for example, one cm square to several cm square or less. The coarsely crushed pieces cut by the coarsely crushing unit **12** pass through a pipe **2** via a chute **9** and are transported to the defibration unit **20**.

(18) The defibration unit **20** defibrates the coarsely crushed material cut by the coarsely crushing unit **12**. Specifically, the defibration unit **20** defibrates the raw material cut by the coarsely crushing unit **12** to generate a defibrated product. Here, “defibrate” means to disentangle a raw material in which a plurality of fibers is bound together into individual fibers. The defibration unit **20** has a function of separating substances such as resin particles, ink, toner, and anti-bleeding agent adhering to the raw material from the fibers.

(19) The material that has passed through the defibration unit **20** is referred to as the defibrated product. The defibrated product may include, in addition to the disentangled defibrated product fibers, resin particles separated from the fibers when the fibers are disentangled, that is, resin particles for binding a plurality of fibers together, color materials such as ink, toner, and the like, additives such anti-bleeding agents and paper strength enhancers. The defibrated product that has been disentangled has a string-like or flat string-like shape. The disentangled defibrated product may be in a state in which it is not entangled with other disentangled fibers, that is, in an independent state, or may be in a state in which it is entangled with other disentangled defibrated products to form a massive form, that is, in a state of forming lumps.

(20) The defibration unit **20** performs fibrillation in a dry manner. Here, a process such as defibration performed not in a liquid but in air such as the atmosphere is referred to as a dry process. The defibration unit **20** includes the impeller mill, for example. Specifically, the defibration unit **20** includes a rotor that rotates at high speed and a liner that is positioned at the outer periphery of the rotor. The coarsely crushed pieces cut by the coarsely crushing unit **12** are pinched between the rotor and the liner of the defibration unit **20** and defibrated. The defibration unit **20** generates an airflow by rotating the rotor. With this airflow, the defibration unit **20** can suck the coarsely crushed pieces, which are raw materials, from the pipe **2** through an introduction port **22** and transport the defibrated product to a discharge port **24**. The defibrated product is sent out from the discharge port **24** to a pipe **3** and transported to the sorting unit **40** via the pipe **3**. In the present embodiment, the sheet manufacturing apparatus **1** includes a defibrating blower **26** as an airflow generating device, and the defibrated product is transported to the sorting unit **40** by the airflow generated by the defibrating blower **26**.

(21) The sorting unit **40** is provided with an introduction port **42** into which the defibrated product disentangled by the defibration unit **20** flows from the pipe **3** together with the airflow. The sorting unit **40** sorts the defibrated products introduced from the introduction port **42** according to the fiber length. More specifically, the sorting unit **40** sorts the defibrated products defibrated by the defibration unit **20** into a defibrated product having a predetermined size or less as a first sorted product, and a defibrated product, as a second sorted product, larger than the first sorted product. The first sorted product includes fibers or particles, and the like, and the second sorted product includes, for example, large fibers, undefibrated pieces, coarsely crushed pieces that are not sufficiently defibrated, and lumps in which fibrillated fibers are aggregated or entangled.

(22) The sorting unit **40** includes, for example, a drum unit **41** and a housing unit **43** that accommodates the drum unit **41**.

(23) The drum unit **41** is a cylindrical sieve that is rotationally driven by a motor. The drum unit **41** includes a mesh and functions as a sieve. According to the mesh openings, the drum unit **41** sorts the defibrated products into a first sorted product smaller than the size of the mesh opening and a second sorted product larger than the size of the mesh opening. Examples of the mesh of the drum unit **41** can include a wire mesh, an expanded metal obtained by stretching a metal plate with cuts, or a punching metal obtained by forming holes in a metal plate using a press or the like.

(24) The defibrated product introduced from the introduction port **42** is sent into the drum unit **41** along with the airflow, and the rotation of the drum unit **41** causes the first sorted product to drop downward from the mesh openings of the drum unit **41**. The second sorted product that cannot pass through the mesh openings of the drum unit **41** is flowed by the airflow flowing into the drum unit **41** from the introduction port **42**, guided to the discharge port **44**, and delivered to a pipe **8**. The pipe **8** couples the inside of the drum unit **41** and the pipe **2**. The second sorted product flowing through the pipe **8** is returned to the defibration unit **20** and defibrated.

(25) The first sorted product sorted by the drum unit **41** passes through the mesh openings of the drum unit **41**, is dispersed in the air, and descend toward a mesh belt **46** of the first web forming unit **45** located below the drum unit **41**.

(26) The first web forming unit **45** includes the mesh belt **46**, rollers **47** and a suction unit **48**. The mesh belt **46** is an endless belt, suspended by the three rollers **47** and rotated by the movement of the rollers **47**. The surface of the mesh belt **46** is composed of a mesh with openings of a predetermined size. Among the first sorted products descending from the sorting unit **40**, fine particles each having a size that can pass through the mesh opening fall below the mesh belt **46**, and fibers each having a size that cannot pass through the mesh opening are accumulated on the mesh belt **46**, and transported along with the mesh belt **46** in the direction of the arrow. The fine particles that fall from the mesh belt **46** includes particles that are relatively small and have low-density in the defibrated product (first sorted product), that is, resin particles that are unnecessary for binding between fibers, color materials, and additives, and are removal materials that are not used when the sheet manufacturing apparatus **1** manufactures the sheet S.

(27) The mesh belt **46** moves at a constant speed $V1$ during the normal operation for manufacturing the sheet S. Here, “during the normal operation” means during operation except during execution of start control and stop control of the sheet manufacturing apparatus **1**. More specifically, it refers to the time when the sheet manufacturing apparatus **1** is manufacturing the sheets S with desired quality.

(28) The suction unit **48** sucks air from below the mesh belt **46**. The suction unit **48** is coupled to a dust collecting unit **27** via a pipe **23**. The dust collecting unit **27** is a filter-type or cyclone-type dust collector that separates fine particles from the airflow. A collection blower **28** is installed downstream of the dust collecting unit **27**, and the collection blower **28** functions as a dust collection suction unit that sucks air from the dust collecting unit **27**. Also, the air discharged by the collection blower **28** is discharged outside the sheet manufacturing apparatus **1** through a pipe **29**.

(29) In the transport path of the mesh belt **46**, mist-containing air is supplied by the humidifying unit **210** downstream of the sorting unit **40**. The mist, which is fine particles of water generated by the humidifying unit **210**, descends toward a first web W1 to supply moisture to the first web W1. As a result, the amount of moisture contained in the first web W1 is adjusted, and adsorption of fibers to the mesh belt **46** due to static electricity can be suppressed.

(30) The sheet manufacturing apparatus **1** includes the rotating body **49** that cuts the first web W1 accumulated on the mesh belt **46**. The first web W1 is separated from the mesh belt **46** at the position where the mesh belt **46** is folded back by a roller **47** and divided by the rotating body **49**.

(31) The rotating body **49** includes a rotation blade shape that has plate-like blades and rotates. The rotating body **49** is disposed at a position where the first web W1 separated from the mesh belt **46** and the blade come into contact with each other. As the rotating body **49** rotates, for example, in the direction indicated by the arrow R in the figure, the first web W1 separated from the mesh belt **46**

and transported collides with the blades and is cut into subdivided bodies P. The subdivided bodies P divided by the rotating body **49** descend inside a pipe **7** and are transported to the mixing unit **50** by the airflow flowing inside the pipe **7**.

(32) The mixing unit **50** includes an additive supply unit **52** that supplies an additive containing resin, a pipe **54** that communicates with the pipe **7** and through which an airflow containing the subdivided bodies P flows, and a mixing blower **56**. The mixing unit **50** mixes the fiber composing the subdivided body P with an additive containing a resin.

(33) In the mixing unit **50**, an airflow is generated by the mixing blower **56**, and the subdivided bodies P and the additives are transported while being mixed in the pipe **54**. Further, the subdivided bodies P are disentangled in the course of flowing inside the pipe **7** and the pipe **54** and become finer fibers.

(34) The additive supply unit **52** is coupled to an additive cartridge (not shown) that accumulates additives, and supplies the additive inside the additive cartridge into the pipe **54**. The additive supply unit **52** stores an additive made of fine powder or fine particles inside the additive cartridge. The additive supply unit **52** includes a discharge unit **52a** that delivers the stored additive to the pipe **54**.

(35) The additive supplied by the additive supply unit **52** contains resin for binding the plurality of fibers. The resin contained in the additive is a thermoplastic resin or a thermosetting resin, and examples thereof include AS resin, ABS resin, polypropylene, polyethylene, polyvinyl chloride, polystyrene, acrylic resin, polyester resin, polyethylene terephthalate, polyphenylene ether, polybutylene terephthalate, polyamide, polycarbonate, polyacetal, polyphenylene sulfide, and polyetheretherketone.

(36) These resins may be used singly or as an appropriate mixture. That is, the additive may contain a single substance, or may be a mixture, and may contain a plurality of types of particles, each of which is composed of a single substance or a plurality of substances. Moreover, the additive may be fibrous or powdery.

(37) The resin contained in the additive is melted by heating to bind a plurality of fibers together. Therefore, in a state in which the resin is mixed with the fibers and the resin is not heated to a temperature at which the resin melts, the fibers are not bound together.

(38) Due to the airflow generated by the mixing blower **56**, the subdivided bodies P descending through the pipe **7** and the additive supplied by the additive supply unit **52** are sucked into the pipe **54** and pass through the mixing blower **56**. Due to the action of the airflow generated by the mixing blower **56** and the rotating part such as the blades of the mixing blower **56**, the fibers that make up the subdivided bodies P and the additive are mixed, and this mixture, that is, the mixture of the first sorted product and the additive, is transported through the pipe **54** to the accumulation unit **60**.

(39) The accumulation unit **60** includes a second web forming unit **70**, introduces the mixture of the first sorted product and the additive that has passed through the mixing unit **50** from an introduction port **62**, and disentangles the entangled defibrated product (first sorted product), and causes the product to fall while dispersing it in the air. Furthermore, when the additive resin supplied from the additive supply unit **52** is fibrous, the accumulation unit **60** disentangles the entangled resin. As a result, the accumulation unit **60** can accumulate the mixture on the second web forming unit **70** with good uniformity.

(40) The accumulation unit **60** includes a drum unit **61** and a housing unit **63** that accommodates the drum unit **61**. The drum unit **61** is a cylindrical sieve that is rotationally driven by a motor. The drum unit **61** includes a mesh and functions as a sieve. Due to this mesh opening, the drum unit **61** allows fibers and particles smaller than the mesh opening to pass through the mesh and descend from the drum unit **61**. The configuration of the drum unit **61** is the same as that of the drum unit **41**, for example.

(41) The second web forming unit **70** is disposed below the drum unit **61**. The second web forming unit **70** accumulates the materials that has passed through the accumulation unit **60** to form a

second web W2 (corresponding to a web). The second web forming unit 70 includes, for example, a mesh belt 72, rollers 74 and a suction mechanism 76.

(42) The mesh belt 72 is an endless belt, suspended by a plurality of rollers 74 and rotated by the movement of the rollers 74. The mesh belt 72 is made of, for example, metal, resin, cloth, non-woven fabric, or the like. The surface of the mesh belt 72 is composed of a mesh with openings of a predetermined size. Among the fibers and particles falling from the drum unit 61, fine particles each having a size that can pass through the mesh opening fall below the mesh belt 72, and fibers each having a size that cannot pass through the mesh opening accumulate on the mesh belt 72, and are transported along with the mesh belt 72 in the direction of the arrow. The mesh belt 72 moves at a constant speed V2 during the normal operation of manufacturing the sheet S.

(43) The mesh opening of the mesh belt 72 is fine and can be of a size that does not allow most of the fibers and particles falling from the drum unit 61 to pass through the mesh.

(44) A suction mechanism 76 is provided below the mesh belt 72. The suction mechanism 76 includes a suction blower 77, and the suction force of the suction blower 77 can generate a downward airflow in the suction mechanism 76.

(45) The suction mechanism 76 sucks the mixture dispersed in the air by the accumulation unit 60 onto the mesh belt 72. As a result, the formation of the second web W2 on the mesh belt 72 can be promoted, and the discharge speed from the accumulation unit 60 can be increased. Furthermore, the suction mechanism 76 can form a downflow in the drop path of the mixture, and can prevent the defibrated product and additives from becoming entangled during the drop.

(46) As described above, the second web W2 that contains a large amount of air and that is soft and swollen is formed by passing through the accumulation unit 60. The second web W2 accumulated on the mesh belt 72 is transported to the sheet forming unit 80.

(47) Air containing mist is supplied by a humidifying unit 212 downstream of the accumulation unit 60 in the transport path of the mesh belt 72. As a result, the mist generated by the humidifying unit 212 is supplied to the second web W2, and the amount of moisture contained in the second web W2 is adjusted. As a result, adsorption of fibers to the mesh belt 72 due to static electricity can be suppressed.

(48) The sheet manufacturing apparatus 1 includes the transport unit 79 that transports the second web W2 on the mesh belt 72 to the sheet forming unit 80. The transport unit 79 includes, for example, a mesh belt 79a, rollers 79b, and a suction mechanism 79c.

(49) The suction mechanism 79c includes a blower (not shown), and generates upward airflow for the mesh belt 79a by the suction force of the blower. This airflow sucks the second web W2, and the second web W2 separates from the mesh belt 72 and is attracted to the mesh belt 79a. The mesh belt 79a is moved by the rotation of the rollers 79b and transports the second web W2 to the sheet forming unit 80.

(50) In this way, the transport unit 79 separates the second web W2 formed at the mesh belt 72 from the mesh belt 72 and transports the second web W2.

(51) The sheet forming unit 80 forms the sheet S by performing at least one of heating and compression (pressurization) on the second web W2. The sheet forming unit 80 of the present embodiment forms the sheet S by pressurizing and heating the second web W2 accumulated on the mesh belt 72 and transported by the transport unit 79. The sheet forming unit 80 applies heat to the fibers and additives contained in the second web W2 to bind the plurality of fibers in the mixture to each other through the resin.

(52) The sheet forming unit 80 includes a pressurizing unit 82 that pressurizes the second web W2 and a heating unit 84 that heats the second web W2 pressurized by the pressurizing unit 82.

(53) The pressurizing unit 82 is composed of a pair of calender rollers 85, and pinches and pressurizes the second web W2 with a predetermined nip pressure. The thickness of the second web W2 is reduced (thinner) by being pressurized, so that the density of the second web W2 is increased. One of the pair of calender rollers 85 is a drive roller driven by a motor (not shown), and

the other is a driven roller. The calender rollers **85** are rotated by the driving force of the motor, and transports the second web W2, which has been increased in density by pressurizing, to the heating unit **84**.

(54) The heating unit **84** includes, for example, a heating roller, a hot press molding machine, a hot plate, a hot air blower, an infrared heater, a flash fixing device, and the like. In the present embodiment, the heating unit **84** includes a pair of heating rollers **86**. The heating rollers **86** are heated to a preset temperature by a heater installed inside or outside. The heating rollers **86** pinch the second web W2 pressurized by the calender rollers **85** and apply heat to the second web W2 to form the sheet S.

(55) One of the pair of heating rollers **86** is a drive roller driven by a motor (not shown), and the other is a driven roller. The heating rollers **86** are rotated by the driving force of the motor and transports the heated sheet S to the cutting unit **90**.

(56) The cutting unit **90** cuts the sheet S formed by the sheet forming unit **80** into a predetermined length. The cutting unit **90** of the present embodiment includes a first cutting unit **92** that cuts the sheet S in a direction intersecting a direction in which the sheet S is transported and a second cutting unit **94** that cuts the sheet S in a direction parallel to the transporting direction. The second cutting unit **94** cuts the sheet S that has passed through the first cutting unit **92**, for example. As a result, a cut sheet S of a predetermined size is formed. The cut sheet S is discharged to a discharge unit **96**. The discharge unit **96** includes a first discharge unit **96a** and a second discharge unit **96b** on which the sheet S can be placed. The first discharge unit **96a** and the second discharge unit **96b** are trays or stackers.

(57) The sheet manufacturing apparatus **1** also includes the detection unit **100** at the exit (downstream) of the sheet forming unit **80**. The detection unit **100** is configured to be able to measure the density information about the sheet S. A detailed configuration of the detection unit **100** will be described later. The controller **300** determines whether the sheet S is acceptable based on the density information, about the sheet S, measured by the detection unit **100**. Then, for example, a sheet Sa that is determined to be acceptable (non-defective product) is discharged to the first discharge unit **96a**, and a sheet Sb that is determined to be unacceptable (defective product) is discharged to the second discharge unit **96b**. By automating the determination of the sheet S, the number of man-hours for determination can be reduced, and the operational efficiency of the sheet manufacturing apparatus **1** can be improved. That is, it is possible to easily determine the sheet S while suppressing labor costs.

(58) Next, the configuration of the detection unit **100** will be described. As shown in FIGS. 2A and 2B, the detection unit **100** includes an optical sensor **110** and a reference member **120**. The optical sensor **110** measures density information about the sheet S formed by the sheet forming unit **80**. The density information about the sheet S includes, for example, the density of each of RGB color components of the color of the sheet S and the RGB color difference of the sheet S. The optical sensor **110** is disposed at the exit (downstream) of the sheet forming unit **80**. The optical sensor **110** of the present embodiment is disposed between the sheet forming unit **80** and the cutting unit **90** in the sheet S transport path. Note that the optical sensor **110** may be disposed between the cutting unit **90** and the discharge unit **96**. The optical sensor **110** can be disposed according to the layout of respective units of the sheet manufacturing apparatus **1**.

(59) The optical sensor **110** has a light emitting unit **111** capable of emitting light and an RGB sensor **112** as a light receiving unit capable of receiving light. The light emitting unit **111** is, for example, a white light emitting diode (LED) and emits light toward the sheet S. The RGB sensor **112** includes three color sensors capable of receiving respective wavelengths of red, green, and blue, and receives light reflected by the sheet S. Each color sensor of the RGB sensor **112** acquires a voltage value proportional to the amount of received light. Based on each voltage value, the density of each of RGB color components of the color of the sheet S and the RGB color difference of the sheet S can be acquired. As a result, it is possible to reliably determine whether the sheet S is

acceptable based on the density (whiteness) of the sheet S and the color difference of the sheet S. Further, by individually setting the density (whiteness) and each color difference specification of the sheet S, the sheet S having the quality desired by the user can be determined.

(60) The optical sensor **110** of the present embodiment is disposed above the transported sheet S, and the light emitting unit **111** and the RGB sensor **112** are disposed to face the upper face S1 of the sheet S. That is, since the light emitting surface of the light emitting unit **111** and the light receiving face of the RGB sensor **112** face downward, adhesion of foreign matter such as paper dust to the light emitting unit **111** and the RGB sensor **112** is suppressed, and accurate measurement can be performed.

(61) The reference member **120** is a member used for calibration of the optical sensor **110** and the like, and is used for adjusting the sensitivity of the RGB sensor **112** in the optical sensor **110**, for example. The reference member **120** is provided at a position facing the optical sensor **110**. The reference member **120** of the present embodiment is disposed below the optical sensor **110**, and the optical sensor **110** and the reference member **120** are disposed facing each other with the transported sheet S interposed therebetween during a time when the sheet manufacturing apparatus **1** is operating.

(62) The reference member **120** has a reference color portion **121**. The reference color portion **121** is provided on a flat face, of the reference member **120**, that can face the optical sensor **110**. The reference color portion **121** is a formation film on which a reference color (color/density) having a constant reflectance with respect to light emitted from the light emitting unit **111** is printed. When performing calibration of the optical sensor **110**, the illuminance of the light emitting unit **111** is adjusted based on the light emitted by the light emitting unit **111** to the reference color portion **121** and a voltage value (AD value) acquired by the light received by the RGB sensor **112**, so that the sensitivity of the RGB sensor **112** is adjusted. As a result, the performance of the optical sensor **110** can be maintained. In addition, individual variation of the optical sensor **110** can be suppressed. In addition, it is possible to determine whether the mounting state of the optical sensor **110** is good or bad. Note that the reference color portion **121** can appropriately set a reference color (color/density) for defining acceptance/unacceptance determination of the formed sheet S. That is, it is possible to set the reference color (color/density) to a reflectance that is a threshold value for determining acceptance/unacceptance of the sheet S. As a result, it is possible to determine the sheet S having the density (whiteness) and color difference desired by the user.

(63) The reference member **120** is rotatably held. The reference member **120** has a rotation shaft **125**. The rotation shaft **125** is rotated by the driving force of the motor, and the entire reference member **120** rotates as the rotation shaft **125** rotates. Specifically, the reference member **120** rotates between a first state CD1 (FIG. 2A) in which the reference color portion **121** faces the optical sensor **110** (light emitting unit **111**, RGB sensor **112**) when the optical sensor **110** is calibrated and a second state CD2 (FIG. 2B) in which the reference color portion **121** faces away from the optical sensor **110** during a time when the sheet manufacturing apparatus **1** is operating. The controller **300** drives and controls the rotation direction of the motor to displace the reference member **120** to the first state CD1 or the second state CD2.

(64) The detection unit **100** also includes a photointerrupter **127** that determines whether the reference member **120** is in the first state CD1 or the second state CD2. The photointerrupter **127** is a sensor that includes a light emitting unit and a light receiving unit facing each other, and determines the presence or absence of an object by detecting that the object blocks the light emitted from the light emitting unit by the light receiving unit. A protrusion **126** is formed at a portion of the reference member **120**. When the reference member **120** rotates and the light receiving unit detects that light from the light emitting unit of the photointerrupter **127** is blocked by the protrusion **126**, it is determined that reference member **120** is in the first state CD1 (not in the second state CD2). Here, the positions of the protrusion **126** and the photointerrupter **127** are set so that the reference color portion **121** faces the optical sensor **110** when the protrusion **126** is detected

by the photointerrupter **127**. The photointerrupters **127** may be disposed on both sides of the reference member **120** so that the first state **CD1** and the second state **CD2** of the reference member **120** can be detected.

(65) In the first state **CD1**, the optical sensor **110** (light emitting unit **111**, RGB sensor **112**) faces the reference color portion **121**, so that the optical sensor **110** can be calibrated. On the other hand, in the second state **CD2**, since the reference color portion **121** faces downward, adhesion of foreign matter such as paper dust to the reference color portion **121** is suppressed. Further, by driving the rotation shaft **125**, the reference member **120** can be easily displaced between the first state **CD1** and the second state **CD2**. Moreover, a space for displacing the reference member **120** between the first state **CD1** and the second state **CD2** is not required, and an increase in the size of the sheet manufacturing apparatus **1** can be suppressed.

(66) Further, a distance **D1** between the surface of the reference color portion **121** and the optical sensor **110** when the reference color portion **121** faces the optical sensor **110** (first state **CD1**) is equal to a distance **D2** between the sheet **S** transported when the reference member **120** is in the second state **CD2** and the sheet manufacturing apparatus **1** is in operation and the optical sensor **110**. As a result, the optical sensor **110** is calibrated under the environment same as that when measuring the density information about the formed sheet **S**, so the density information about the sheet **S** can be accurately measured.

(67) Next, a method for controlling the sheet manufacturing apparatus **1** will be described. In the present embodiment, a control method for the detection unit **100** will be described. As shown in FIG. **3**, in the sheet manufacturing apparatus **1**, a calibration process of the optical sensor **110** is first executed (step **S100**). The controller **300** controls the optical sensor **110** and the reference member **120** to adjust the sensitivity of the optical sensor **110** (RGB sensor **112**). Next, a forming process for the sheet **S** is executed (step **S200**). The controller **300** controls the supply unit **10** and the like to form the sheet **S**. Next, a discrimination process for the formed sheet **S** is executed (step **S300**). The controller **300** determines whether the sheets **S** are acceptable based on the density information about the sheets **S** acquired by the optical sensor **110**, and sorts the sheets into the acceptable sheet **Sa** (non-defective product) and the unacceptable sheet **Sb** (defective product). Note that the calibration process may be performed while the sheet **S** forming process is being performed. That is, the calibration process of the optical sensor **110** may be executed before the sheet **S** is transported to the detection unit **100**. Further, the determination process for the sheet **S** may be performed while the sheet **S** forming process is being performed. In this way, the sheet manufacturing apparatus **1** can be efficiently operated. Each process will be described below. Note that the sheet **S** forming process (step **S200**) is the same as the content when describing the configuration of the sheet manufacturing apparatus **1**, so that description thereof will be omitted.

(68) FIGS. **4A**, **4B**, **4C**, and **4D** show details of the calibration process (step **S100**) for the optical sensor **110**. First, the controller **300** acquires an AD value of the RGB sensor **112** (each color sensor) when the light emitting unit **111** does not emit light, and determines whether the RGB sensor **112** is abnormal in the initial stage. Specifically, first, in step **S101**, the controller **300** sets the reference member **120** to the first state **CD1**. For example, when the reference member **120** is in the second state **CD2**, the controller **300** rotates the rotation shaft **125** so that the reference member **120** is in the first state **CD1**.

(69) Next, in step **S102**, controller **300** determines whether the reference member **120** is in first state **CD1** (FIG. **2A**). Specifically, it is determined whether the photointerrupter **127** has detected the protrusion **126**. When it is determined that the reference member **120** is in the first state **CD1** (YES), the process proceeds to step **S103**. On the other hand, when it is determined that the reference member **120** is not in the first state **CD1** (NO), the process proceeds to step **S106**.

(70) When the process proceeds to step **S106**, the controller **300** determines whether the number of times that it is determined that the reference member **120** is not in the first state **CD1** is a predetermined number of times. The predetermined number of times is, for example, the second

time. In this case, when the number of times that it is determined that the reference member **120** is not in the first state **CD1** is the first time (NO), the process proceeds to step **S101**. On the other hand, when the number of times that it is determined that the reference member **120** is not in the first state **CD1** is the second time (YES), the process proceeds to step **S107**. In step **S107**, the abnormality process is executed. For example, when the reference member **120** cannot be set to the first state **CD1**, it is assumed that there is an abnormality in the photointerrupter **127**, an attachment failure of the reference member **120**, a motor failure, or the like.

(71) When the process proceeds to step **S103**, the controller **300** drives only the RGB sensor **112** without causing the light emitting unit **111** to emit light. Then, the voltage values acquired by the RGB sensor **112** are digitally converted to acquire the AD values of respective color sensors.

(72) Next, in step **S104**, the controller **300** determines whether the acquired AD value is equal to or less than a predetermined value. The predetermined value is an abnormality determination threshold value of an AD value when the light emitting unit **111** does not emit light. When the light emitting unit **111** does not emit light, the amount of light received from the reference color portion **121** is smaller than that when the light emitting unit **111** emits light, so that when the AD value of each color sensor indicates a low value (equal to or less than the abnormality determination threshold value), the controller **300** determines that the value is normal (YES), and the process proceeds to step **S105** to store the AD value of each color sensor in the memory when the light emitting unit **111** does not emit light. On the other hand, when the AD value is larger than the abnormality determination threshold value (NO), the process proceeds to step **S107**. In step **S107**, the abnormality process is executed. In this case, it is presumed that the RGB sensor **112** itself is abnormal.

(73) Next, the controller **300** acquires the AD value of the RGB sensor **112** (each color sensor) when the light emitting unit **111** emits light, and adjusts the sensitivity of the RGB sensor **112**. Specifically, in step **S108**, the controller **300** causes the light emitting unit **111** to emit light using the initially set an initial light emission value parameter.

(74) Next, in step **S109**, after the light emission amount of the light emitting unit **111** is stabilized, the controller **300** acquires the AD value of the color sensor with the highest or lowest sensitivity among the color sensors of the RGB sensor **112**. In the present embodiment, the AD value of the R (red) color sensor having the highest sensitivity is acquired. By acquiring the AD value of the color sensor with the highest sensitivity or the lowest sensitivity, it is possible to efficiently adjust the illuminance of the light emitting unit **111**.

(75) Next, in step **S110**, the controller **300** determines whether the acquired AD value is within a predetermined value range. Specifically, it is determined whether the AD value acquired by the initial light emission value parameter is equal to or greater than the lower limit of the abnormality determination threshold value and equal to or less than the upper limit of the abnormality determination threshold value. That is, it is determined whether the color sensor in the initial state is acceptable. Then, when it is determined that the acquired AD value is within the predetermined value range (YES), the process proceeds to step **S111**. On the other hand, when it is determined that the acquired AD value is not within the predetermined value range (NO), the process proceeds to step **S117**, the optical sensor **110** is determined to be abnormal, and the abnormality process is performed.

(76) Next, in step **S111**, the controller **300** calculates the ratio between the AD value acquired in step **S109** (in the present embodiment, the R (red) color sensor) and the target AD value of the color sensor. As a result, the difference (ratio) between the AD value in the initial state and the target AD value is calculated.

(77) Next, in step **S112**, based on the ratio calculated in step **S111**, the controller **300** reflects the result in the light emission value parameter of the light emitting unit **111**. For example, when the ratio of the AD value in the initial state to the target AD value is 1:2, the parameter is changed to a new light emission value parameter corresponding to twice the initial light emission value

parameter of the light emitting unit **111**. In this example, the adjustment is made so that the amount of light emitted from the light emitting unit **111** increases (brightening). As a result, the sensitivity of the optical sensor **110** is adjusted, and the AD value of the color sensor corresponding to the target value can be acquired. Then, the light emitting unit **111** is caused to emit light using the adjusted light emission value parameter.

(78) Next, in step **S113**, after the light emission amount of the light emitting unit **111** is stabilized, the controller **300** acquires the AD value of the color sensor with the highest or lowest sensitivity among the color sensors of the RGB sensor **112**. In the present embodiment, the AD value of the R (red) color sensor having the highest sensitivity is acquired.

(79) Next, in step **S114**, the controller **300** determines whether the acquired AD value is within the target value range. Specifically, it is determined whether the acquired AD value is within the allowable range of the target value set in step **S111**. When it is determined that the acquired AD value is within the range of the target value (YES), the process proceeds to step **S115**. In step **S115**, the controller **300** stores the adjusted light emission value parameter in the memory. On the other hand, when it is determined that the acquired AD value is not within the range of the target value (NO), the process proceeds to step **S116**. In step **S116**, the controller **300** determines whether the number of times that it is determined that the acquired AD value is not within the range of the target value is equal to or less than a predetermined number of times. When it is determined that the number of times is equal to or less than the predetermined number of times (YES), the process proceeds to step **S111**. On the other hand, when it is determined that the number of times is greater than the predetermined number of times (NO), the process proceeds to step **S117**, the optical sensor **110** is determined to be abnormal, and the abnormality process is executed.

(80) Next, the controller **300** acquires the AD values of the RGB sensor **112** after sensitivity adjustment, and calculates a relational expression (proportional expression) between the density (whiteness) and the AD values. Specifically, in step **S118**, the controller **300** causes the light emitting unit **111** to emit light using the adjusted light emission value parameters stored in the memory, and acquires the AD value of each color sensor in the RGB sensor **112** after the light emission amount of the light emitting unit **111** is stabilized.

(81) Next, in step **S119**, the controller **300** stores the acquired AD value of each color sensor in the memory.

(82) Next, in step **S120**, the controller **300** calculates a proportional expression (linear expression) using an AD value of each color sensor acquired when the light emitting unit **111** is not emitting light (step **S105**) and an AD value of each color sensor acquired when the light emitting unit **111** is emitting light (step **S115**). As a result, as shown in FIG. 4D, the relational expression (proportional expression) between density (whiteness) and AD value is calculated for each color sensor.

(83) Next, in step **S121**, the controller **300** stores the calculated proportional expression of each color sensor in the RGB sensor **112** in the memory. With the above, the calibration process of the optical sensor **110** is completed.

(84) Next, the determination process for the sheet S (step **S300**) will be described. FIGS. 5A, 5B, and 5C show the details of the sheet S determination process. First, the controller **300** executes the process of forming the sheet S, and determines whether the formed sheet S has reached the measurable area of the optical sensor **110**. Specifically, in step **S301**, the controller **300** sets the reference member **120** to the second state CD2. For example, when the reference member **120** is in the first state CD1, the rotation shaft **125** is rotated so that the reference member **120** is in the second state CD2.

(85) Next, in step **S302**, controller **300** determines whether the reference member **120** is in second state CD2 (FIG. 2B). Specifically, it is determined whether the protrusion **126** of the photointerrupter **127** is detected. When it is determined that the reference member **120** is in the second state CD2 (YES), the process proceeds to step **S303**. On the other hand, when it is determined that the reference member **120** is not in the second state CD2 (NO), the process

proceeds to step S307.

(86) When the process proceeds to step S307, the controller **300** determines whether the number of times that it is determined that the reference member **120** is not in the second state CD2 is a predetermined number of times. The predetermined number of times is, for example, the second time. In this case, when the number of times that it is determined that the reference member **120** is not in the second state CD2 is the first time (NO), the process proceeds to step S301. On the other hand, when the number of times that it is determined that the reference member **120** is not in the second state CD2 is the second time (YES), the process proceeds to step S308. In step S308, the abnormality process is executed. For example, when the reference member **120** cannot be set to the second state CD2, it is assumed that there is an abnormality in the photointerrupter **127**, an attachment failure of the reference member **120**, a motor failure, or the like.

(87) In step S303, the controller **300** determines whether the formed sheet S has been transported in the measurable area of the optical sensor **110**. For example, the light emitting unit **111** of the optical sensor **110** emits light, and the RGB sensor **112** receives the reflected light. Since the amount of received light changes depending on whether there is reflection from the sheet S, it is possible to detect whether the sheet S has been transported. When it is determined that the sheet S has been transported (YES), the process proceeds to step S304. On the other hand, when it is determined that the sheet S is not transported (NO), the process proceeds to step S302.

(88) Next, in step S304, the controller **300** acquires AD values of the RGB sensor **112** at a predetermined cycle (for example, one msec). Specifically, the AD value of each color sensor is acquired. Note that the predetermined cycle can be appropriately set according to the sheet S transporting speed.

(89) Next, in step S305, the controller **300** averages the acquired AD values of each color sensors. Specifically, the average value of the predetermined number of AD values sequentially acquired of each color sensor is calculated.

(90) Next, in step S306, the controller **300** determines whether the average AD value of each color sensor calculated in step S305 is equal to or greater than a specified value. The specified value is a threshold value for determining the presence or absence of the sheet S in the measurable area of the optical sensor **110**. When it is determined that the average AD value of each color sensor is equal to or greater than the specified value (YES), that is, when the sheet S reaches the measurable area of the optical sensor **110**, the process proceeds to step S309. On the other hand, when it is determined that the average AD value of each color sensor is less than the specified value (NO), the process proceeds to step S305.

(91) After step S309, the controller **300** calculates the RGB density of the sheet S and the RGB color difference of the sheet S based on the AD values acquired from the optical sensor **110**, and determines whether the sheet S is acceptable based on the calculation result. Specifically, in step S309, the controller **300** acquires AD values of the RGB sensor **112** at the predetermined cycle (for example, one msec). Specifically, a plurality of AD values of each color sensor is acquired. Note that when the sheet S reaches the measurable area of the optical sensor **110**, the thickness dimension of the leading end portion of the sheet S tends to vary. Therefore, acquisition of AD values by the RGB sensor **112** is started after a predetermined time has elapsed. As a result, the AD values can be acquired accurately.

(92) Next, in step S310, the controller **300** calculates an average value, a deviation value, a maximum value, and a minimum value for each color sensor using a predetermined number of AD values sequentially acquired as the sheet S is transported.

(93) Next, in step S311, the controller **300** calculates an average value using only AD values within a predetermined deviation value for each color sensor. As a result, for example, by excluding AD values obtained by measuring peculiar portions such as black or white lumps formed at the sheet S, variations in calculation results can be reduced.

(94) Next, in step S312, the controller **300** calculates the density (whiteness) of each color sensor

based on the proportional expression (FIG. 4D) using the average value calculated in step S311. Specifically, the density corresponding to the AD value of the R color sensor, the density corresponding to the AD value of the G color sensor, and the density corresponding to the AD value of the B color sensor are calculated.

(95) Next, in step S313, the controller 300 averages the densities of respective color sensors calculated from the proportional expression in step S312. Specifically, the sum of the density of the R color sensor, the density of the G color sensor, and the density of the B color sensor is divided by three. As a result, the RGB density (whiteness) of the color of the sheet S is calculated.

(96) Next, in step S314, the controller 300 calculates the difference (difference in density) between the maximum value and the minimum value of the density calculation values between the color sensors calculated from the proportional expression in step S312. For example, the difference between the density (maximum value) calculated from the AD value of the R color sensor and the density (minimum value) calculated from the AD value of the B color sensor is calculated. From this, the RGB color difference of the sheet S is calculated.

(97) Next, in step S315, the controller 300 determines whether the average density calculated in step S313 is equal to or greater than a predetermined value. The predetermined value in this step is a density determination threshold value DS1 (FIG. 5C) of the sheet S desired by the user. Then, when it is determined that the average density is equal to or higher than the predetermined value (YES), the process proceeds to step S316. On the other hand, when it is determined that the average concentration is less than the predetermined value (NO), the process proceeds to step S318.

(98) When the process proceeds to step S318, the sheet S for which the average density is determined to be less than the predetermined value is determined to be unacceptable (defective product), and the unacceptable sheet Sb is discharged to the second discharge unit 96b.

(99) When the process proceeds to step S316, the controller 300 determines whether the difference in density calculated in step S314 is equal to or less than a predetermined value. The predetermined value in this step is an RGB color difference determination threshold value IS1 of the sheet S (FIG. 5C). When it is determined that the difference in density is equal to or less than the predetermined value (YES), the process proceeds to step S317. On the other hand, when it is determined that the difference in density is greater than the predetermined value (NO), the process proceeds to step S318. When the process proceeds to step S318, the sheet S for which the difference in density is determined to be greater than the predetermined value is determined to be unacceptable (defective product), and the unacceptable sheet Sb is discharged to the second discharge unit 96b.

(100) When the process proceeds to step S317, the sheet S for which the difference in density is determined to be equal to or less than a predetermined value is determined to be acceptable (non-defective product), and the acceptable sheet Sa is discharged to the first discharge unit 96a. That is, as shown in FIG. 5C, the acceptable sheet Sa has a density (whiteness) equal to or greater than the density determination threshold value DS1 and a color difference equal to or less than the determination threshold value IS1. On the other hand, the unacceptable sheet Sb has a density (whiteness) less than the density determination threshold value DS1 or a color difference greater than the determination threshold value IS1.

(101) According to the present embodiment, it is possible to easily determine the sheet S (sheet Sa or sheet Sb) based on the density information (whiteness/color difference) about the formed sheet S. Further, when the recycled sheet S is repeatedly recycled, the density of the sheet S decreases, the lengths of the fibers composing the sheet S decreases, the adhesive force between the fibers decreases, and the strength of the sheet S decreases. According to the present embodiment, by determining the sheet S based on the density information about the sheet S, it is also possible to determine whether the sheet S satisfies a predetermined strength. By individually setting threshold values of the density (whiteness) and the color difference that define the determination of the sheet S, it is possible to easily determine the sheet S having the quality desired by the user.

Claims

1. A used paper recycling apparatus comprising: a defibration unit that defibrates used paper to generate a defibrated product; an accumulation unit that accumulates the defibrated product to form a web; a sheet forming unit that forms a sheet by applying at least one of heating and compression to the web; a cutting unit that cuts the sheet into a predetermined length; and an optical sensor provided at an exit of the sheet forming unit and including three color sensors that receive wavelengths of red, green, and blue, respectively, the optical sensor measuring density information about the sheet, the density information including a density of each of RGB color components of a color of the sheet and an RGB color difference of the sheet, the RGB color difference being a difference between a maximum value and a minimum value among density calculation values that correspond to average voltage values for the three color sensors, respectively, the average voltage values being average values of voltage values obtained for a predetermined cycle at the three color sensors, respectively.
 2. The used paper recycling apparatus according to claim 1, wherein the optical sensor is disposed between the sheet forming unit and the cutting unit.
 3. The used paper recycling apparatus according to claim 1, further comprising: a reference member provided at a position facing the optical sensor, wherein the used paper recycling apparatus calibrates the optical sensor based on a result of measuring a reference color portion provided on the reference member.
 4. The used paper recycling apparatus according to claim 3, wherein the reference member is rotatably held, and the reference member rotates so that the reference color portion faces the optical sensor when the calibration is performed, and the reference color portion faces away from the optical sensor during a time when the used paper recycling apparatus is operating.
 5. The used paper recycling apparatus according to claim 4, wherein a distance between a surface of the reference color portion and the optical sensor when the reference color portion faces the optical sensor is equal to a distance between the sheet and the optical sensor when the used paper recycling apparatus is in operation.
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